Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

SPARKvue: Motion Sensors

Match Graph lab • PASCO wireless motion sensor

1. Open SPARKvue

Take out your **laptop**, open **Google Chrome**, and go to:

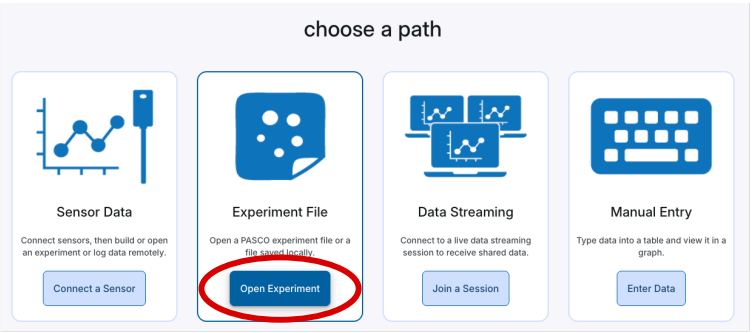
sparkvue.pasco.com

2. Download the lab file

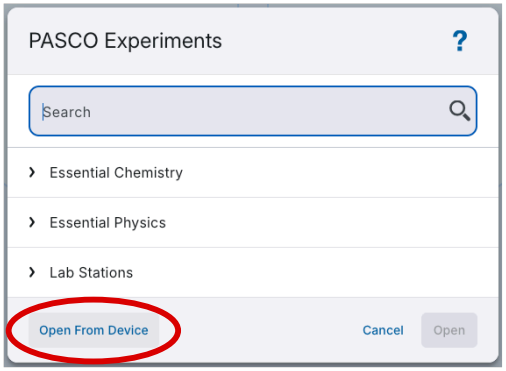
On your **class page**, click **Match Graph lab file**. It saves to your **Downloads** folder.

3. Open the lab file in SPARKvue

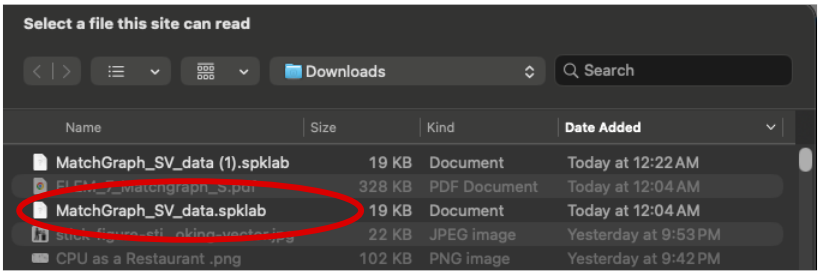
1. On the **choose a path** screen, under **Experiment File**, click **Open Experiment** (a)
2. In the **PASCO Experiments** box, click **Open From Device**, bottom left (b)
3. Open your **Downloads** folder, click **MatchGraph**, then click **Open** (c)



a. Open Experiment

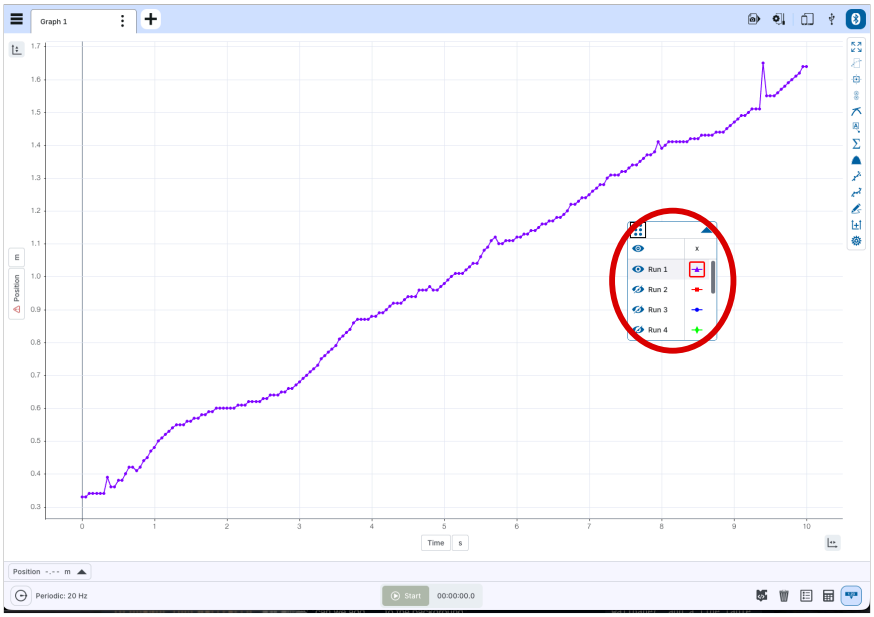


b. Open From Device



c. Downloads → MatchGraph → Open

Check: SPARKvue opens a graph like the one below. **Run 1** is showing; the other runs are hidden. **Runs 1 to 9** are the nine target graphs, drawn in *The nine runs*.



d. The lab file, open. The ringed box is the **legend**: click an **eye** to show or hide a run.

4. Connect the sensor

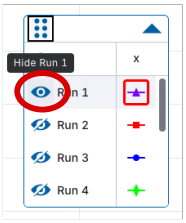
1. Put the sensor on a table, its **round screen at waist height**, facing **3–4 m** of clear floor.
2. Press the **power button**. The light **blinks red**.
3. In SPARKvue, click **Bluetooth** and pick the number printed on your sensor's label.

Check: the light now **blinks green**.

5. Match a run

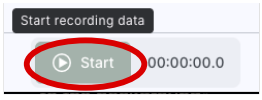
The sensor measures **your distance from its round screen**. Walk **away** and the line goes **up**; walk **towards** it and the line goes **down**. Stay at least **0.3 m** away.

1. **Show one run.** Have **only one of Runs 1–9 visible at a time**. Click the **eye** beside a run to **hide** or **show** it. Start with **Run 1** showing and every other run hidden.



Open eye: the run is **showing**.
Crossed-out eye: the run is **hidden**.
Here Run 1 is showing; Runs 2–4 are hidden.

2. **Stand** where the run's line starts.
3. **Walk it.** Your partner clicks **Start** (*Start recording data*), bottom of the screen; it records for **10 s**. Your walk appears in the legend as a new run.



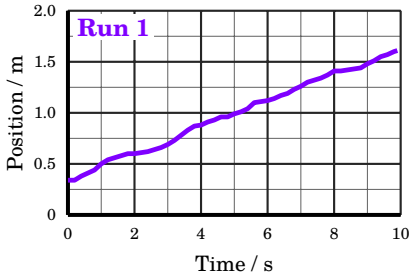
4. **Compare and retry** until your line follows the target run. Hide each old attempt with its eye, so the screen shows the target and your newest walk.
5. **Write** how you moved under that run in *The nine runs*, and tick *Matched?*

Check: everyone walks at least two runs. **Care:** keep the floor clear, and walk.

The nine runs

Words to use: stationary • constant speed • away from • towards • faster • slower • speeding up • slowing down • turn round

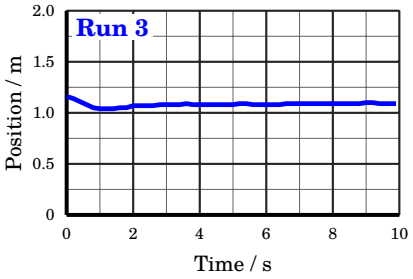
Runs 1–9 from PASCO's MatchGraph! lab for SPARKvue, redrawn from the lab file. Each run is drawn in its colour in the SPARKvue legend.



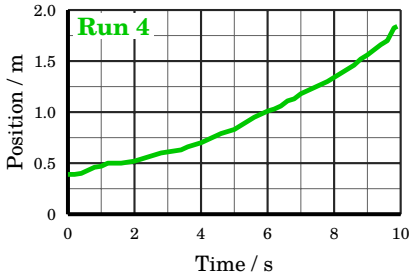
How I moved: **Matched?** ☐



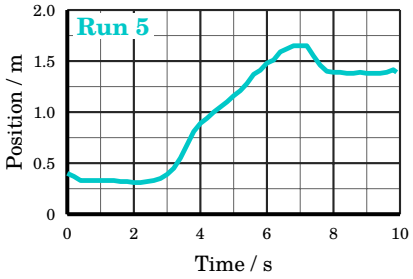
How I moved: **Matched?** ☐



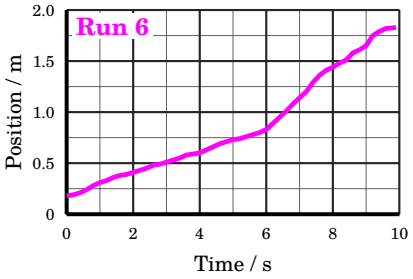
How I moved: **Matched?** ☐



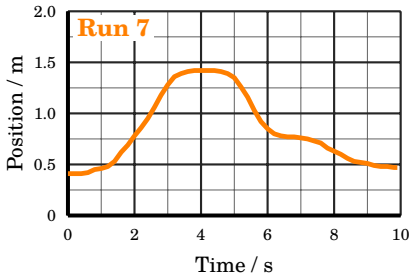
How I moved: **Matched?** ☐



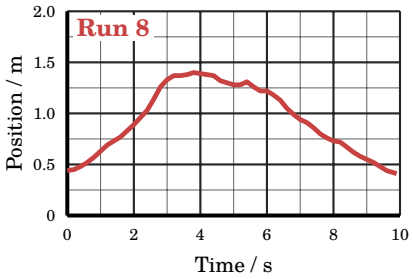
How I moved: **Matched?** ☐



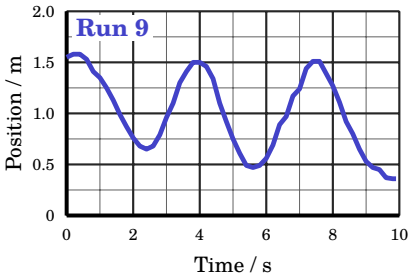
How I moved: **Matched?** ☐



How I moved: **Matched?** ☐



How I moved: **Matched?** ☐



How I moved: **Matched?** ☐

Questions and analysis

1. Which run shows a **constant speed** the whole way? Run _____

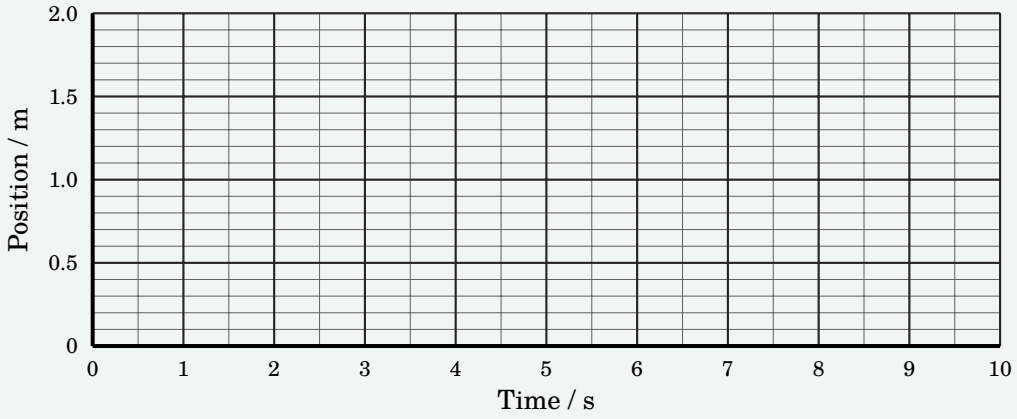
2. Which run shows someone **at rest** the whole time? Run _____

3. Which run shows moving **away** and **speeding up**? Run _____

4. Which run was the hardest to match? Use your description to explain what made it hard.

To go further — design your own run

Sketch a target run of your own on these axes: at least three parts, and every part something a person can walk. Swap sheets with another group. Can they match it first time?



Check: no part of your line is vertical, and it never goes below 0.3 m.